

Functions

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1) Functions

Functions are a special type of equation that relates the value of one variable, namely **dependent variable (DV)**, with the value of one or more other variables, named **independent variables (IVs)**. Which variable is going to be put as the dependent and which is the independent is in principle arbitrary. In its explicit form, which is the one more often found, the function is an equation stating the value of the variable elected as dependent, which appears alone in one side of the equation, using an expression containing the IVs, which appears on the other side. Notably, it is also possible for a function to have more than one expression, with each expression holding for a given specific interval of values the IV can admit: this type of function is called piecewise function. In mathematical language, the symbolism for a function is $f(IV_1, \dots, IV_n)$, where the IVs are distinguished from each other by the subscript in this case and $f(IV_1, \dots, IV_n)$ refers to the mathematical expression underlying a given function. Since the value given by this expression is also the value of the DV, one can write $DV = f(IV_1, \dots, IV_n)$. For functions having 1 IV, it is common to have the DV symbolized by y and the IV by x , case in which the symbolism becomes $f(x)$.

Formally, for an equation to be a function, a set of criteria exist. Firstly there must not be more than one possible result of the DV for a same combination of values of the IVs. Secondly, and a bit more too formal in nature, the set of values assigned to be possible for the IV(s), which must accompany the equation (e.g. $IVs \in \text{real numbers}$), must contain only combinations of the IVs for which there is a possible value for the DV in hand. **Therefore, if there are IV(s) values that lead to an undefined value of the DV, these combinations of IVs are not part**

of the set of values assigned to be possible for the IV(s). On what concerns nomenclature, one calls **domain** the set of values that the IVs can assume, and **image** the set of values that the DV is in principle free to assume. **Codomain** is the subset of the image containing only the values that indeed are admitted by the DV given the domain and the expression relating DV and IVs. If the domain is not specified alongside the function, it is assumed to be all real numbers.

In the context of functions, it may be worth remembering, for a complete picture so far, that any mathematical expression we can define by coupling mathematical operations imply a mapping (or algebraic pattern/logic) between the input (domain) and output (codomain). Furthermore, as any equation relating two or more variables, unless the domain is quite restrictive, the amount of solutions is infinite. Note that if the domain consists of all real numbers, being infinite is about not only the direction of growth and decline, but also within any interval delimited by upper and lower limits: this is a consequence of the indefinite amount of decimal places a real number can have. This makes it impossible to name the successor or predecessor of a number.

Particularly in the context of functions having more than one IV (called here multi-IV function), prior knowledge of the given context may be useful to verify that the IVs are indeed independent. If the variables functioning as IVs in a function are really independent, they are able to vary freely. This means it is not possible or even plausible to try to establish a mathematical relationship between them that maps possible combinations: such a relationship should not exist. If the IVs are seen to do not vary freely, the variables cannot form truly what consists a set of IVs. Moreover, this would mean that, in principle, the expression of the function cannot be used without further information as the one quantitative relationship the occurring values of the variables respect: it is necessary to know further relationship between the so-called “IVs” to know what indeed is the domain.

2) Equivalent and semi-equivalent functions

As discussed previously, just like the algorithm of a specific mathematical operation implies a mapping between a set of inputs and a set of outputs, **coupling mathematical operations sequentially into an expression also yields a specific mapping of its own.** In the context of functions in their explicit form, the value adopted by a variable can be regarded as an input, the coupling of mathematical operations that the input(s) must undergo within the statement defines the mapping and the result (i.e. value of the dependent variable) is the output. In this framework, a notable thing to realize is that a same mapping (or algebraic pattern), as defined in a previous document, can be described by different combinations of mathematical operations (i.e. mathematical expressions). In other words, it is possible to have more than one mathematical expression, formulated with different sets of operations, that can establish the same algebraic pattern. This can be regarded as something that simply “happens to happen” within the scope of the definition of our mathematical operations. The notion of equivalency had in fact already been discussed and shown when solving equations with one unknown in the previous document. As when solving

equations, one possible way to determine or find equivalent expressions is to be mindful about the definition (and thus algorithm) inherent of each mathematical operation defined.

Practical example 1: example of equivalent functions could be the functions $f(x) = 2x/3$ and $f(x) = 4x/6$. Despite fundamentally consisting of different commands (i.e., the numbers x needs to be divided and multiplied by in the two cases are different), the algebraic pattern linking input and output is the same in the two cases.

Here in the context of functions, it is particularly important to beware of punctual exceptions to equivalency. Different functions defined by different expressions may not share entirely, but only part, the algebraic pattern. In this case, they cannot be said to be equivalent, but if the sole reason preventing a full equivalency of their algebraic pattern from happening is the difference in output for a finite number of IV values, these functions can be regarded as **semi-equivalent**.

Practical example 2: Consider the functions given by equations (1), (2) and (3):

$$y = \frac{x^2 + 4x + 4}{(x + 2)} \quad (1)$$

$$y = \frac{(x + 2)^2}{(x + 2)} \quad (2)$$

$$y = x + 2 \quad (3)$$

These functions are all written having in mind operations that could lead to equivalency between them. Equations (1) and (2) are equivalent because the numerator of the expression associated to equation (1) can be rewritten as $(x + 2)^2$ without altering the result for whichever value admitted by x . This can be verified by doing the square of $(x + 2)$ as indicated. Additionally, in the scope of the division operation involved in the expressions of functions given by equations (1) and (2), the outcome will also remain the same for each and every value of x used (the denominator is after all the same), with $x = -2$ being the one value of x for which the function is undefined.

The division operation is the very reason why the functions in equation (3) differs from the ones in equations (1) and (2). Despite describing the same underlying algebraic pattern as the other two equations, the output in equation (3) for $x = -2$ is defined. The equivalency between equation (3) and the other two is true for whichever other value of x , because in the right hand side of equations (1) and (2) there is going to simply be a division and a multiplication by a same number: this is the same as never having operated an input provided that the division is not by 0. The input $x = -2$ aside, equations (1), (2) and (3) are

equivalent. But since as a whole we cannot say they share the same solutions, they can, at best, be said to be "semi-equivalent".